

Technology Career Roles Guide

A plain-English reference describing what each technology job does, the skills it needs, how to enter it, typical certifications and salary ranges. Written as a knowledge source for a career question-answering chatbot.

Role: AI Engineer

In one line: An AI Engineer builds software products that use artificial intelligence models to solve real problems.

What does an AI Engineer do?

An AI Engineer takes AI models (often large language models or vision models) and turns them into working applications. The AI Engineer does not usually invent new algorithms. Instead, the AI Engineer connects models to data, writes the code around them, and makes sure the system is fast, safe and reliable in production.

Daily work of an AI Engineer

- An AI Engineer writes Python code to call and control AI models.
- An AI Engineer builds RAG pipelines so a chatbot can answer from company documents.
- An AI Engineer writes and tests prompts, then measures the quality of the answers.
- An AI Engineer deploys models as APIs and monitors cost, speed and errors.
- An AI Engineer works with product managers to decide what the AI feature should do.

Skills required for an AI Engineer

- Python programming is the core skill of an AI Engineer.
- An AI Engineer needs to understand machine learning basics: training, inference, embeddings, overfitting.
- An AI Engineer should know LLM tools such as LangChain, LlamaIndex, Ollama, Hugging Face Transformers.
- An AI Engineer needs vector databases like FAISS, Chroma, Pinecone or Qdrant.
- An AI Engineer needs APIs and backend skills: REST, FastAPI or Flask, JSON.
- An AI Engineer needs cloud and deployment skills: Docker, Git, AWS or Azure or GCP.
- Soft skills for an AI Engineer: clear communication, problem breakdown, reading research papers.

How to become an AI Engineer

To become an AI Engineer: first learn Python well, then learn data handling with pandas and NumPy, then learn machine learning basics, then learn deep learning and transformers, then build 3 to 5 real projects such as a RAG chatbot, a document summariser and a recommendation app, and finally learn deployment with Docker and a cloud provider.

A common entry path to AI Engineer is starting as a Python developer, data analyst, or junior ML engineer and moving across after building an AI project portfolio.

Certifications for an AI Engineer

Useful certifications for an AI Engineer include AWS Certified Machine Learning, Microsoft Azure AI Engineer Associate (AI-102), Google Professional Machine Learning Engineer, and DeepLearning.AI specialisations.

Salary of an AI Engineer

Salary for an AI Engineer in India is roughly 6 to 12 lakh rupees per year at entry level, 12 to 25 lakh at mid level, and 30 lakh and above at senior level. In the United States an AI Engineer typically earns about 110,000 to 180,000 US dollars per year. These are rough ranges and change with city, company and year.

Related roles and who it suits

Related job titles to AI Engineer: Machine Learning Engineer, LLM Engineer, Applied Scientist, GenAI Developer, MLOps Engineer.

The AI Engineer role suits someone who likes building products, enjoys experimenting, and is comfortable when there is no single correct answer.

Role: Machine Learning Engineer

In one line: A Machine Learning Engineer trains machine learning models and puts them into production systems.

What does a Machine Learning Engineer do?

A Machine Learning Engineer prepares data, trains and tunes models, evaluates their accuracy, and deploys them so that other software can use them. A Machine Learning Engineer sits between a Data Scientist, who explores and experiments, and a Software Engineer, who ships reliable code.

Daily work of a Machine Learning Engineer

- A Machine Learning Engineer cleans and prepares training data.
- A Machine Learning Engineer trains models and tunes hyperparameters.
- A Machine Learning Engineer measures accuracy, precision, recall and F1 score.
- A Machine Learning Engineer builds retraining pipelines and watches for model drift.
- A Machine Learning Engineer packages models into APIs for other teams.

Skills required for a Machine Learning Engineer

- A Machine Learning Engineer needs Python, scikit-learn, PyTorch or TensorFlow.
- A Machine Learning Engineer needs SQL and data pipeline tools such as Spark or Airflow.
- A Machine Learning Engineer needs maths basics: statistics, probability, linear algebra.
- A Machine Learning Engineer needs MLOps tools: MLflow, Docker, Kubernetes, CI/CD.

How to become a Machine Learning Engineer

To become a Machine Learning Engineer: learn Python and SQL, then statistics, then classical machine learning with scikit-learn, then deep learning, then MLOps and deployment. Build projects that include the full pipeline from raw data to a deployed API.

Many Machine Learning Engineers start as data analysts, software engineers or research interns.

Certifications for a Machine Learning Engineer

Useful certifications for a Machine Learning Engineer: Google Professional Machine Learning Engineer, AWS Certified Machine Learning Specialty, TensorFlow Developer Certificate.

Salary of a Machine Learning Engineer

Salary for a Machine Learning Engineer in India is roughly 6 to 14 lakh rupees per year at entry level and 20 to 40 lakh at senior level. In the United States it is typically 120,000 to 190,000 US dollars per year.

Related roles and who it suits

Related job titles: AI Engineer, Data Scientist, MLOps Engineer, Applied Scientist.

The Machine Learning Engineer role suits someone who enjoys maths, experiments and measurable results.

Role: Data Scientist

In one line: A Data Scientist studies data to find patterns and helps a business make better decisions.

What does a Data Scientist do?

A Data Scientist asks questions of data, runs statistical analysis, builds predictive models, and explains the findings to business teams using charts and clear language. A Data Scientist spends a lot of time cleaning data and communicating results, not only building models.

Daily work of a Data Scientist

- A Data Scientist writes SQL queries to pull data from databases.
- A Data Scientist explores data in Jupyter notebooks with pandas and matplotlib.
- A Data Scientist runs A/B tests and explains whether a change actually helped.
- A Data Scientist builds dashboards and presents insights to managers.

Skills required for a Data Scientist

- A Data Scientist needs Python or R, pandas, NumPy and visualisation libraries.
- A Data Scientist needs strong SQL.
- A Data Scientist needs statistics: hypothesis testing, regression, probability distributions.
- A Data Scientist needs business understanding and storytelling with data.

How to become a Data Scientist

To become a Data Scientist: learn SQL and Excel, then Python with pandas, then statistics, then machine learning, then build case-study projects that answer a business question end to end.

A common entry path to Data Scientist is starting as a Data Analyst or Business Analyst.

Certifications for a Data Scientist

Useful certifications for a Data Scientist: Google Data Analytics Certificate, Microsoft Certified Azure Data Scientist Associate (DP-100), IBM Data Science Professional Certificate.

Salary of a Data Scientist

Salary for a Data Scientist in India is roughly 5 to 12 lakh rupees per year at entry level and 18 to 35 lakh at senior level. In the United States it is typically 100,000 to 165,000 US dollars per year.

Related roles and who it suits

Related job titles: Data Analyst, Machine Learning Engineer, Quantitative Analyst, Decision Scientist.

The Data Scientist role suits curious people who like statistics and explaining things clearly.

Role: Data Analyst

In one line: A Data Analyst turns raw data into reports and dashboards that a business can act on.

What does a Data Analyst do?

A Data Analyst collects data, cleans it, analyses it and presents it. A Data Analyst answers questions such as which product sold best last quarter or why customers stopped buying. A Data Analyst is one of the easiest entry points into a technology career.

Daily work of a Data Analyst

- A Data Analyst writes SQL queries and builds Excel models.
- A Data Analyst creates dashboards in Power BI or Tableau.
- A Data Analyst checks data quality and fixes errors in reports.
- A Data Analyst presents weekly numbers to business teams.

Skills required for a Data Analyst

- A Data Analyst needs Excel or Google Sheets at an advanced level.
- A Data Analyst needs SQL, which is the single most important skill for this role.
- A Data Analyst needs a BI tool such as Power BI, Tableau or Looker.
- A Data Analyst benefits from basic Python with pandas.

How to become a Data Analyst

To become a Data Analyst: learn Excel, then SQL, then one BI tool, then basic statistics, then build 3 dashboard projects using public datasets.

Data Analyst is often a first job with no prior experience, and it is a common stepping stone to Data Scientist or Analytics Engineer.

Certifications for a Data Analyst

Useful certifications for a Data Analyst: Google Data Analytics Professional Certificate, Microsoft Power BI Data Analyst (PL-300), Tableau Desktop Specialist.

Salary of a Data Analyst

Salary for a Data Analyst in India is roughly 3.5 to 8 lakh rupees per year at entry level and 10 to 20 lakh at senior level. In the United States it is typically 65,000 to 100,000 US dollars per year.

Related roles and who it suits

Related job titles: Business Analyst, Reporting Analyst, Analytics Engineer, MIS Executive.

The Data Analyst role suits detail-oriented people who like organising information.

Role: Network Engineer

In one line: A Network Engineer designs, builds and maintains the computer networks that connect devices and offices.

What does a Network Engineer do?

A Network Engineer makes sure data moves reliably and securely between computers, servers, and the internet. A Network Engineer configures routers, switches, firewalls and VPNs, plans IP addressing, and fixes problems when the network is slow or down.

Daily work of a Network Engineer

- A Network Engineer configures routers and switches using command line interfaces.
- A Network Engineer troubleshoots connectivity problems with tools like ping, traceroute and Wireshark.
- A Network Engineer plans IP address schemes and VLANs for offices.
- A Network Engineer sets up firewalls and VPNs for secure remote access.
- A Network Engineer monitors bandwidth, uptime and network alerts.

Skills required for a Network Engineer

- A Network Engineer must understand the OSI model and the TCP/IP model.
- A Network Engineer needs routing and switching knowledge: OSPF, BGP, VLAN, STP.
- A Network Engineer needs IP addressing and subnetting skills, including IPv4 and IPv6.
- A Network Engineer needs network security basics: firewalls, ACLs, NAT, VPN.
- A Network Engineer benefits from automation with Python and Ansible.
- A Network Engineer increasingly needs cloud networking knowledge such as AWS VPC.

How to become a Network Engineer

To become a Network Engineer: learn networking fundamentals through CompTIA Network+ or CCNA material, practise on simulators such as Cisco Packet Tracer or GNS3, learn subnetting until it is easy, then get hands-on with real devices or a home lab, then add automation and cloud networking.

A common entry path to Network Engineer is starting as an IT Support Engineer, Desktop Support Engineer or Network Operations Centre (NOC) technician.

Certifications for a Network Engineer

Useful certifications for a Network Engineer: CompTIA Network+ for beginners, Cisco CCNA as the industry standard, then Cisco CCNP, Juniper JNCIA, AWS Advanced Networking, and Fortinet NSE for security-focused work.

Salary of a Network Engineer

Salary for a Network Engineer in India is roughly 3 to 7 lakh rupees per year at entry level, 8 to 15 lakh at mid level, and 18 to 30 lakh at senior level. In the United States a Network Engineer typically earns 75,000 to 130,000 US dollars per year.

Related roles and who it suits

Related job titles: Network Administrator, NOC Engineer, Network Security Engineer, Cloud Network Engineer, Systems Engineer.

The Network Engineer role suits methodical people who enjoy troubleshooting and infrastructure rather than writing application code.

Role: Cybersecurity Analyst

In one line: A Cybersecurity Analyst protects an organisation's systems and data from attacks.

What does a Cybersecurity Analyst do?

A Cybersecurity Analyst monitors systems for suspicious activity, investigates alerts, closes security gaps and responds to incidents such as malware or phishing. A Cybersecurity Analyst often works in a Security Operations Centre, also called a SOC.

Daily work of a Cybersecurity Analyst

- A Cybersecurity Analyst reviews alerts in a SIEM tool like Splunk or Microsoft Sentinel.
- A Cybersecurity Analyst investigates whether an alert is a real attack or a false positive.
- A Cybersecurity Analyst runs vulnerability scans and asks teams to patch systems.
- A Cybersecurity Analyst writes incident reports and improves security rules.

Skills required for a Cybersecurity Analyst

- A Cybersecurity Analyst needs networking fundamentals and operating system knowledge, especially Linux and Windows.
- A Cybersecurity Analyst needs SIEM tools, log analysis and threat detection skills.
- A Cybersecurity Analyst needs to understand common attacks: phishing, ransomware, SQL injection, DDoS.
- A Cybersecurity Analyst benefits from scripting in Python or PowerShell.

How to become a Cybersecurity Analyst

To become a Cybersecurity Analyst: learn networking first, then Linux, then security fundamentals through CompTIA Security+, then practise on platforms like TryHackMe or Hack The Box, then target a SOC analyst role.

A common entry path to Cybersecurity Analyst is from IT support or network administration.

Certifications for a Cybersecurity Analyst

Useful certifications for a Cybersecurity Analyst: CompTIA Security+, CompTIA CySA+, Certified Ethical Hacker (CEH), and later CISSP for senior roles.

Salary of a Cybersecurity Analyst

Salary for a Cybersecurity Analyst in India is roughly 4 to 9 lakh rupees per year at entry level and 15 to 30 lakh at senior level. In the United States it is typically 80,000 to 130,000 US dollars per year.

Related roles and who it suits

Related job titles: SOC Analyst, Information Security Analyst, Penetration Tester, Security Engineer.

The Cybersecurity Analyst role suits people who are curious, suspicious by nature and enjoy investigation.

Role: Cloud Engineer

In one line: A Cloud Engineer builds and manages systems that run on cloud platforms like AWS, Azure and Google Cloud.

What does a Cloud Engineer do?

A Cloud Engineer moves applications to the cloud, designs the servers, storage and networks there, controls cost, and automates the setup using code instead of manual clicking.

Daily work of a Cloud Engineer

- A Cloud Engineer creates cloud infrastructure using Terraform or CloudFormation.
- A Cloud Engineer manages virtual machines, containers, storage buckets and databases.
- A Cloud Engineer sets up access permissions and security rules.
- A Cloud Engineer reviews cloud bills and reduces unnecessary cost.

Skills required for a Cloud Engineer

- A Cloud Engineer needs deep knowledge of at least one cloud provider: AWS, Azure or Google Cloud.
- A Cloud Engineer needs Linux, networking and scripting skills.
- A Cloud Engineer needs Infrastructure as Code tools such as Terraform.
- A Cloud Engineer needs containers: Docker and Kubernetes.

How to become a Cloud Engineer

To become a Cloud Engineer: learn Linux and networking, then pick one cloud provider and learn its core services, then learn Terraform and Docker, then build projects such as deploying a web app with a database and automated pipeline.

A common entry path to Cloud Engineer is from system administration, IT support or software development.

Certifications for a Cloud Engineer

Useful certifications for a Cloud Engineer: AWS Certified Solutions Architect Associate, Microsoft Azure Administrator (AZ-104), Google Associate Cloud Engineer, HashiCorp Terraform Associate.

Salary of a Cloud Engineer

Salary for a Cloud Engineer in India is roughly 5 to 10 lakh rupees per year at entry level and 20 to 35 lakh at senior level. In the United States it is typically 100,000 to 155,000 US dollars per year.

Related roles and who it suits

Related job titles: DevOps Engineer, Site Reliability Engineer, Cloud Architect, Platform Engineer.

The Cloud Engineer role suits people who like automation and building reliable infrastructure.

Role: DevOps Engineer

In one line: A DevOps Engineer makes it fast, safe and automatic to release software.

What does a DevOps Engineer do?

A DevOps Engineer connects the development team and the operations team. A DevOps Engineer builds CI/CD pipelines so code is tested and deployed automatically, sets up monitoring, and keeps systems running with minimum downtime.

Daily work of a DevOps Engineer

- A DevOps Engineer builds and fixes CI/CD pipelines in Jenkins, GitHub Actions or GitLab CI.
- A DevOps Engineer writes Dockerfiles and Kubernetes manifests.
- A DevOps Engineer sets up monitoring and alerting with Prometheus and Grafana.
- A DevOps Engineer automates repetitive operations tasks with scripts.

Skills required for a DevOps Engineer

- A DevOps Engineer needs Linux and shell scripting.
- A DevOps Engineer needs Git, CI/CD tools and containerisation.
- A DevOps Engineer needs cloud platforms and Infrastructure as Code.
- A DevOps Engineer needs monitoring and logging tools.

How to become a DevOps Engineer

To become a DevOps Engineer: learn Linux, then Git, then one programming or scripting language, then Docker, then a CI/CD tool, then Kubernetes, then cloud and Terraform.

A common entry path to DevOps Engineer is from software development, system administration or cloud support.

Certifications for a DevOps Engineer

Useful certifications for a DevOps Engineer: AWS Certified DevOps Engineer Professional, Certified Kubernetes Administrator (CKA), Docker Certified Associate.

Salary of a DevOps Engineer

Salary for a DevOps Engineer in India is roughly 5 to 11 lakh rupees per year at entry level and 20 to 40 lakh at senior level. In the United States it is typically 105,000 to 165,000 US dollars per year.

Related roles and who it suits

Related job titles: Site Reliability Engineer, Platform Engineer, Build Engineer, Cloud Engineer.

The DevOps Engineer role suits people who hate repeating manual work and enjoy automating it.

Role: Full Stack Developer

In one line: A Full Stack Developer builds both the front end that users see and the back end that runs behind it.

What does a Full Stack Developer do?

A Full Stack Developer creates the user interface, the server logic and the database connections for a web application. A Full Stack Developer is valuable in small teams and startups because one person can build a complete feature.

Daily work of a Full Stack Developer

- A Full Stack Developer builds pages and components in React or another front-end framework.
- A Full Stack Developer writes API endpoints in Node.js, Django, Spring or FastAPI.
- A Full Stack Developer designs database tables and writes queries.
- A Full Stack Developer fixes bugs reported by users and reviews teammates' code.

Skills required for a Full Stack Developer

- A Full Stack Developer needs HTML, CSS and JavaScript.
- A Full Stack Developer needs a front-end framework such as React, Angular or Vue.
- A Full Stack Developer needs a back-end language such as Node.js, Python or Java.
- A Full Stack Developer needs databases, both SQL like PostgreSQL and NoSQL like MongoDB.
- A Full Stack Developer needs Git and basic deployment skills.

How to become a Full Stack Developer

To become a Full Stack Developer: learn HTML and CSS, then JavaScript, then React, then a backend framework, then databases, then deploy 3 full projects publicly.

Full Stack Developer is a common first job for self-taught programmers and bootcamp graduates because a portfolio matters more than a degree.

Certifications for a Full Stack Developer

Certifications matter less for a Full Stack Developer than a strong GitHub portfolio and live deployed projects.

Salary of a Full Stack Developer

Salary for a Full Stack Developer in India is roughly 4 to 9 lakh rupees per year at entry level and 18 to 35 lakh at senior level. In the United States it is typically 90,000 to 150,000 US dollars per year.

Related roles and who it suits

Related job titles: Front End Developer, Back End Developer, Software Engineer, Web Developer.

The Full Stack Developer role suits people who like seeing a product come together end to end.

Role: Software Engineer

In one line: A Software Engineer designs, writes and maintains computer programs.

What does a Software Engineer do?

A Software Engineer takes a requirement, designs a solution, writes clean and tested code, and keeps improving it. A Software Engineer works in areas such as web, mobile, backend systems, embedded devices or games.

Daily work of a Software Engineer

- A Software Engineer writes and reviews code.
- A Software Engineer writes unit tests and fixes bugs.
- A Software Engineer joins design discussions and daily stand-up meetings.
- A Software Engineer documents how the system works.

Skills required for a Software Engineer

- A Software Engineer needs at least one language deeply: Python, Java, C++, Go or JavaScript.
- A Software Engineer needs data structures and algorithms.
- A Software Engineer needs Git, testing and debugging skills.
- A Software Engineer needs system design knowledge as they grow senior.

How to become a Software Engineer

To become a Software Engineer: pick one language, learn data structures and algorithms, build projects, practise coding interview questions, contribute to open source, then apply for internships or junior roles.

Common entry paths to Software Engineer are a computer science degree, a bootcamp, or self-study with a strong project portfolio.

Certifications for a Software Engineer

Certifications are rarely required for a Software Engineer. Projects, internships and interview performance matter more.

Salary of a Software Engineer

Salary for a Software Engineer in India is roughly 4 to 12 lakh rupees per year at entry level and 20 to 50 lakh at senior level. In the United States it is typically 95,000 to 170,000 US dollars per year.

Related roles and who it suits

Related job titles: Backend Developer, Application Developer, Programmer Analyst, SDE.

The Software Engineer role suits people who enjoy logical problem solving and building things.

Role: Database Administrator

In one line: A Database Administrator keeps databases fast, safe and available.

What does a Database Administrator do?

A Database Administrator, often shortened to DBA, installs and configures database systems, tunes slow queries, plans backups and recovery, and controls who can access which data.

Daily work of a Database Administrator

- A Database Administrator monitors database performance and fixes slow queries.
- A Database Administrator runs and tests backups and restore procedures.
- A Database Administrator manages user permissions and security.
- A Database Administrator plans capacity and upgrades.

Skills required for a Database Administrator

- A Database Administrator needs advanced SQL.
- A Database Administrator needs one database system deeply: Oracle, MySQL, PostgreSQL or SQL Server.
- A Database Administrator needs backup, replication and disaster recovery knowledge.
- A Database Administrator needs Linux and scripting skills.

How to become a Database Administrator

To become a Database Administrator: learn SQL thoroughly, then database internals such as indexes and transactions, then administration tasks on one platform, then backup and high availability.

A common entry path to Database Administrator is from IT support, development or data analysis.

Certifications for a Database Administrator

Useful certifications for a Database Administrator: Oracle Certified Professional, Microsoft Azure Database Administrator (DP-300), AWS Database Specialty.

Salary of a Database Administrator

Salary for a Database Administrator in India is roughly 4 to 8 lakh rupees per year at entry level and 15 to 28 lakh at senior level. In the United States it is typically 85,000 to 135,000 US dollars per year.

Related roles and who it suits

Related job titles: Data Engineer, Database Developer, Site Reliability Engineer.

The Database Administrator role suits careful people who value reliability and order.

Role: Data Engineer

In one line: A Data Engineer builds the pipelines that move and prepare data for analysis.

What does a Data Engineer do?

A Data Engineer collects data from many sources, cleans and transforms it, and stores it in warehouses so analysts and machine learning teams can use it. Without a Data Engineer, a Data Scientist has no reliable data to work with.

Daily work of a Data Engineer

- A Data Engineer writes ETL or ELT pipelines.
- A Data Engineer schedules jobs with Airflow and fixes failed runs.
- A Data Engineer models tables in a data warehouse such as Snowflake or BigQuery.
- A Data Engineer monitors data quality and freshness.

Skills required for a Data Engineer

- A Data Engineer needs strong SQL and Python.
- A Data Engineer needs big data tools such as Spark and Kafka.
- A Data Engineer needs warehouses: Snowflake, BigQuery, Redshift.
- A Data Engineer needs orchestration tools like Airflow and cloud platforms.

How to become a Data Engineer

To become a Data Engineer: learn SQL, then Python, then data modelling, then one cloud warehouse, then Airflow and Spark, then build an end-to-end pipeline project.

A common entry path to Data Engineer is from data analysis, backend development or database administration.

Certifications for a Data Engineer

Useful certifications for a Data Engineer: Google Professional Data Engineer, AWS Certified Data Engineer Associate, Databricks Data Engineer Associate.

Salary of a Data Engineer

Salary for a Data Engineer in India is roughly 5 to 11 lakh rupees per year at entry level and 20 to 40 lakh at senior level. In the United States it is typically 105,000 to 160,000 US dollars per year.

Related roles and who it suits

Related job titles: Analytics Engineer, ETL Developer, Big Data Engineer, Machine Learning Engineer.

The Data Engineer role suits people who like building reliable systems more than making charts.

Role: IT Support Engineer

In one line: An IT Support Engineer helps people fix problems with computers, software and networks.

What does an IT Support Engineer do?

An IT Support Engineer is usually the first person a user contacts when something breaks. An IT Support Engineer resets passwords, sets up laptops, installs software, fixes printer and network issues, and escalates harder problems to specialist teams. IT Support Engineer is one of the most common first jobs in technology.

Daily work of an IT Support Engineer

- An IT Support Engineer answers support tickets and phone calls.
- An IT Support Engineer sets up new employee laptops and accounts.
- An IT Support Engineer troubleshoots Windows, macOS and network issues.
- An IT Support Engineer documents fixes in a knowledge base.

Skills required for an IT Support Engineer

- An IT Support Engineer needs operating system knowledge for Windows and Linux.
- An IT Support Engineer needs basic networking: IP addresses, DNS, Wi-Fi, VPN.
- An IT Support Engineer needs hardware troubleshooting skills.
- An IT Support Engineer needs patience and good communication with non-technical users.

How to become an IT Support Engineer

To become an IT Support Engineer: learn computer hardware and operating systems, then basic networking, then get CompTIA A+ or an equivalent, then apply for helpdesk roles. From there many people move into Network Engineer, Cloud Engineer or Cybersecurity Analyst roles.

IT Support Engineer is usually an entry-level job that does not require a technical degree.

Certifications for an IT Support Engineer

Useful certifications for an IT Support Engineer: CompTIA A+, Google IT Support Professional Certificate, Microsoft 365 Fundamentals.

Salary of an IT Support Engineer

Salary for an IT Support Engineer in India is roughly 2.5 to 5 lakh rupees per year at entry level and 6 to 12 lakh with experience. In the United States it is typically 45,000 to 75,000 US dollars per year.

Related roles and who it suits

Related job titles: Helpdesk Technician, Desktop Support Engineer, System Administrator, NOC Technician.

The IT Support Engineer role suits helpful, patient people who want a practical entry into technology.

Role: QA Engineer

In one line: A QA Engineer tests software to make sure it works correctly before users see it.

What does a QA Engineer do?

A QA Engineer, also called a Quality Assurance Engineer or Test Engineer, writes test cases, runs manual and automated tests, reports bugs, and protects the quality of each release.

Daily work of a QA Engineer

- A QA Engineer writes and runs test cases for new features.
- A QA Engineer builds automated tests with Selenium, Cypress or Playwright.
- A QA Engineer reports bugs clearly with steps to reproduce.
- A QA Engineer runs regression tests before a release.

Skills required for a QA Engineer

- A QA Engineer needs testing fundamentals: test cases, test plans, bug life cycle.
- A QA Engineer needs an automation tool such as Selenium, Cypress or Playwright.
- A QA Engineer needs basic programming in Python, Java or JavaScript.
- A QA Engineer needs API testing skills with tools like Postman.

How to become a QA Engineer

To become a QA Engineer: learn manual testing concepts, then SQL and basic programming, then one automation framework, then API testing, then CI integration.

QA Engineer is a common entry-level role and a good route into software engineering later.

Certifications for a QA Engineer

Useful certifications for a QA Engineer: ISTQB Foundation Level.

Salary of a QA Engineer

Salary for a QA Engineer in India is roughly 3 to 7 lakh rupees per year at entry level and 12 to 25 lakh at senior level. In the United States it is typically 70,000 to 115,000 US dollars per year.

Related roles and who it suits

Related job titles: Test Engineer, SDET, Automation Engineer.

The QA Engineer role suits people who notice details and enjoy breaking things to find weaknesses.